Earned Value Management Dashboard

Construction Economics and Finance · CE725 · Topic 7

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₹14.00 Cr
project budget at completion

19
work packages tracked individually

50 / 50
independent benchmark comparisons agree

The problem, in one number

₹4.02 Cr
PLANNED VALUE
what should have been done

₹3.98 Cr
ACTUAL COST
what it has cost

₹3.47 Cr
EARNED VALUE
what was actually done

Spend is under plan by ₹4 lakh, so the monthly report writes itself as healthy.
The two errors point in opposite directions and cancel.

! **14.7% over budget**
every rupee spent returns 87 paise of budgeted work

! **19 days behind programme**
measured in time, not as a dimensionless index

Budget against spend is two numbers, and two numbers give one comparison. Separating ‘am I doing the work’ from ‘is the work costing what it should’ needs a third measurement: the budgeted cost of the work actually performed.

Three measurements, and comparisons between them

PV — **Budgeted cost of work scheduled**

What should have been done by now?

EV — **Budgeted cost of work performed**

What was actually done?

AC — **Actual cost of doing it**

What did it cost?

Everything else compares two of the three

EV against AC → cost

```
CV = EV − AC
CPI = EV ÷ AC
```

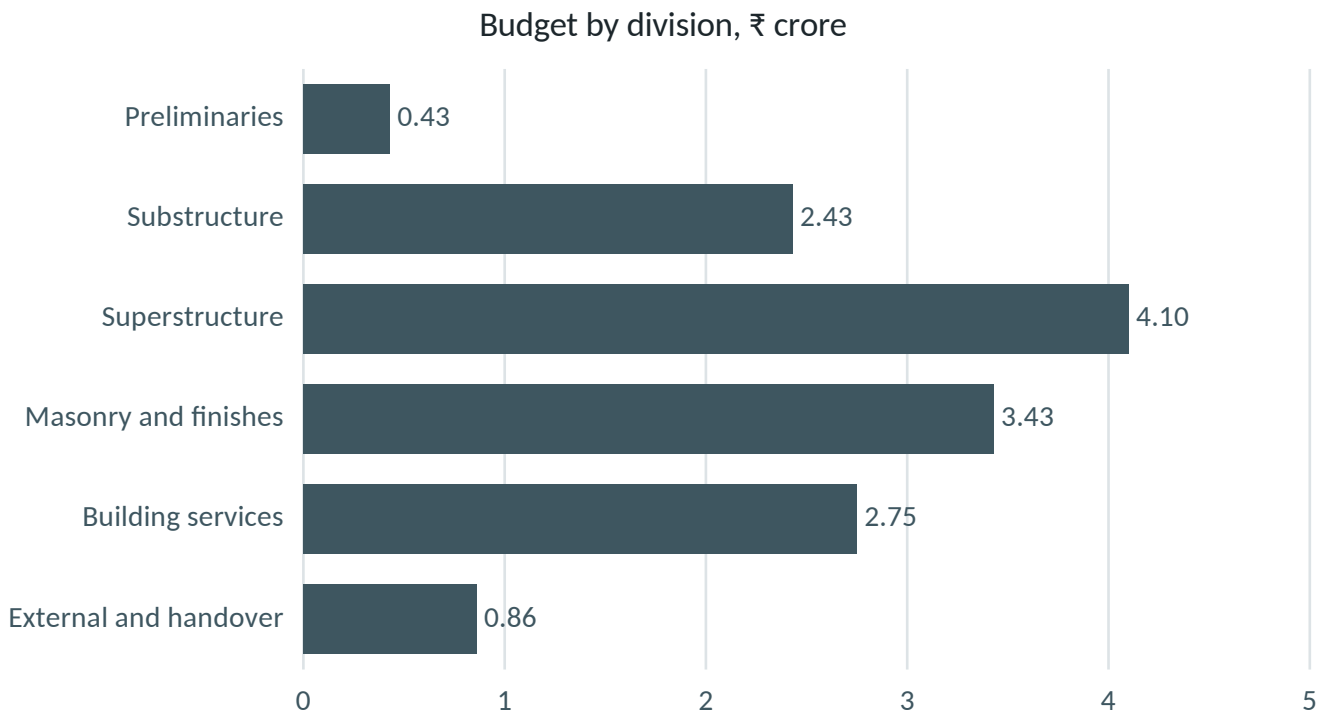
EV against PV → schedule

```
SV = EV − PV
SPI = EV ÷ PV
```

“That is the whole method. The rest is bookkeeping on top of it.”

The tool adds Earned Schedule, four forecasting methods and the same calculation at every work package.

The case study: a G+4 RCC academic block



BAC	₹14.00 crore
Baseline	12 months, Apr 2026 to Mar 2027
Structure	19 leaf work packages, 6 divisions
Data date	end of month 6, Sep 2026

How it went wrong

- Hard rock in excavation — 45% over on that package, two months late
- Cascaded into footings and foundation RCC
- Frame started a month late, running an 18-day slab cycle against 12

! **Synthetic data.** Budgets are order-of-magnitude plausible for an institutional building built to CPWD DSR-style item rates, but they are assumed — not drawn from any tender, estimate or invoice. Slide 12 is about why that choice matters more than it looks.

Four pieces, one repository

1

Python calculation engine

Every EVM measure, computed at work-package level rather than rolled up to a single project curve.

2

Streamlit dashboard

The tool. Verdict and plain-language summary first, then indices, diagnosis and forecast.

3

Live-formula Excel export

Every figure a real formula. Edit a budget or a percent complete and the whole workbook recalculates.

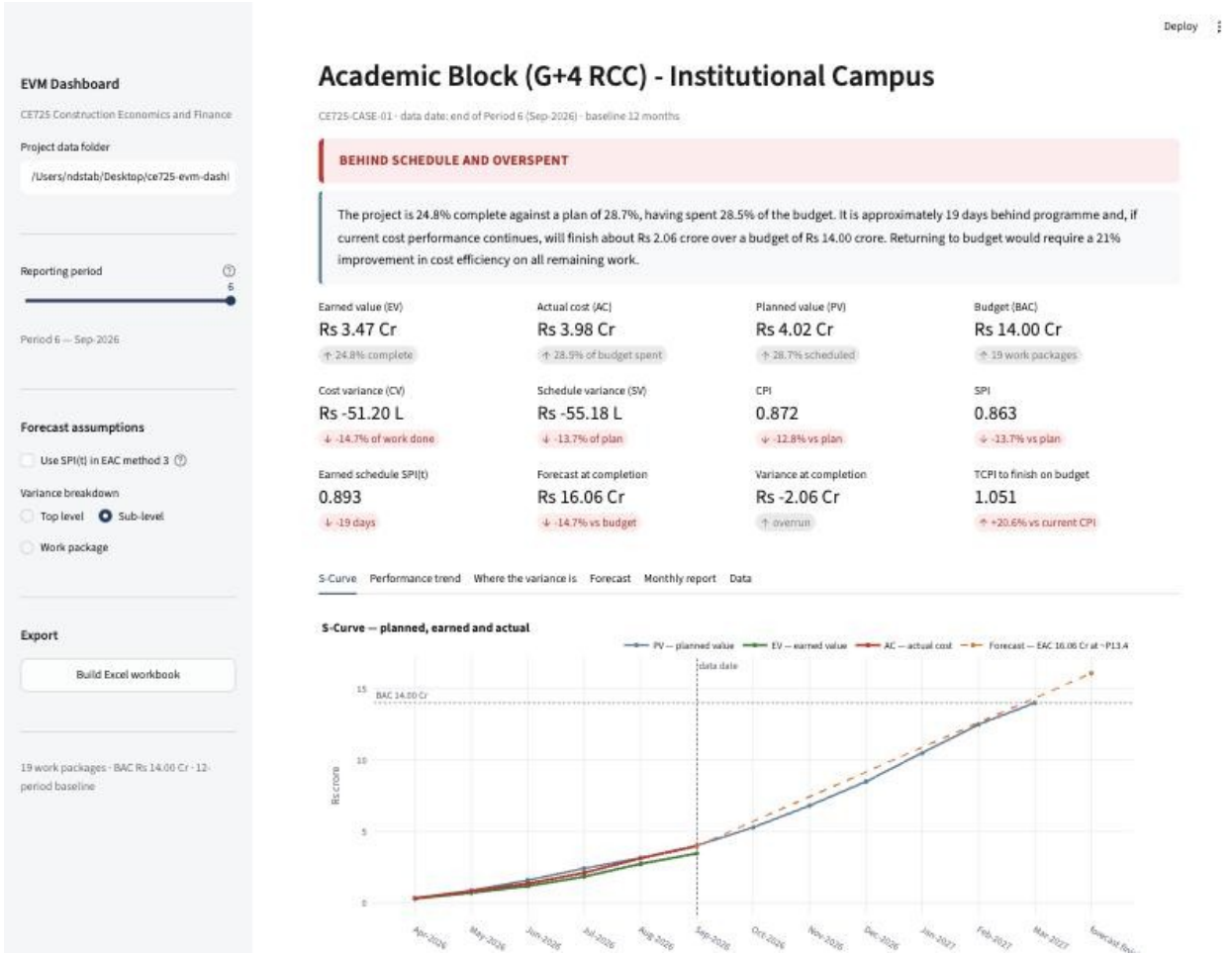
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Independent manual benchmark

A separate Excel workbook, executed not re-implemented. This is how we know the numbers are right.

→ **Switch to the live dashboard** — slides 6 to 9 are driven from the running tool

Where the project stands at month six



CPI 0.872

every rupee spent returns 87 paise of budgeted work

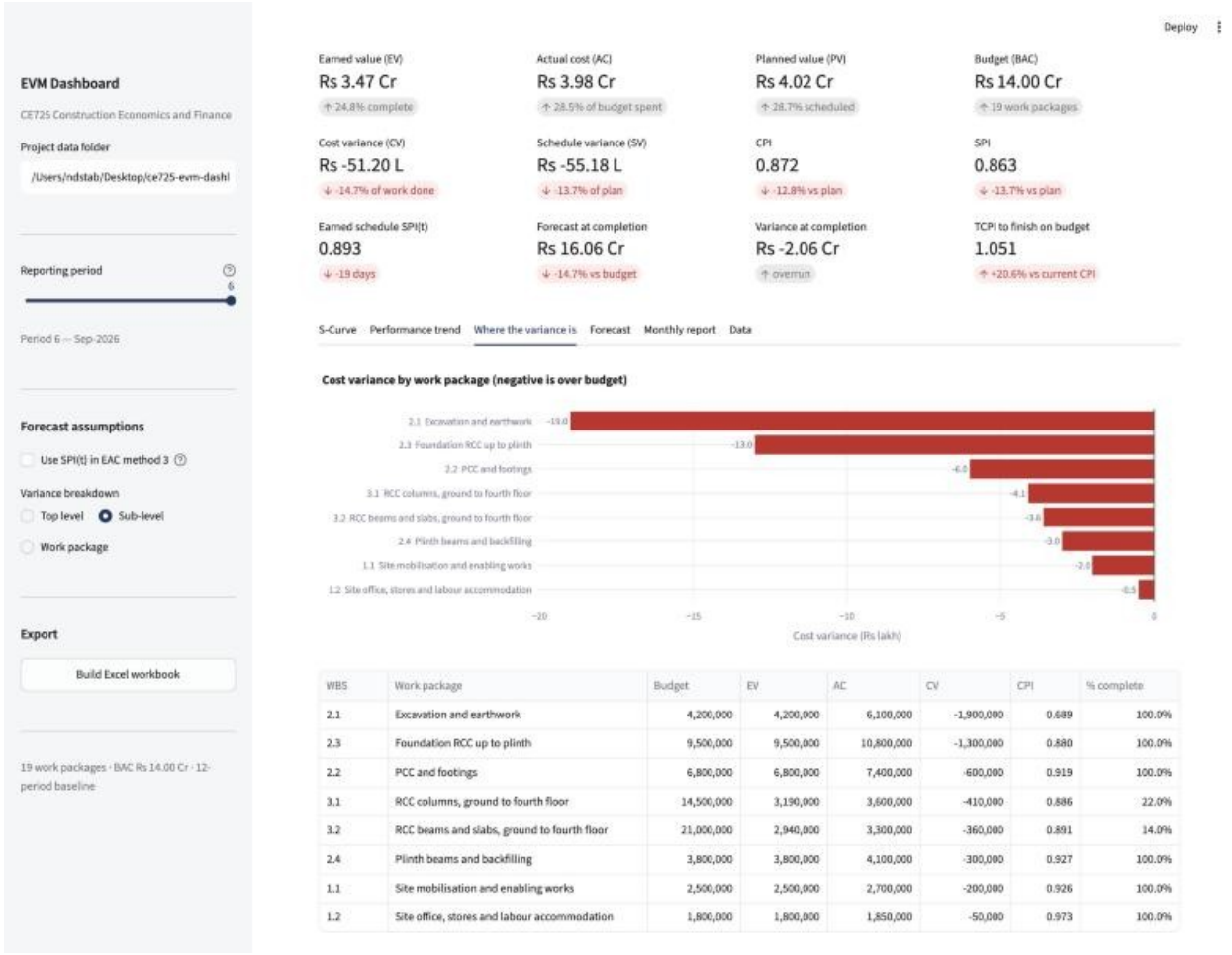
TCPI 1.051

efficiency the remaining work must now achieve

We have sustained 0.872 for six months. TCPI asks for a 20.6% improvement on a level of performance never once achieved.

The budget is gone. What remains is deciding who absorbs it.

Where the variance actually is



WBS	Work package	Budget	EV	AC	CV	CPI	% complete
2.1	Excavation and earthwork	4,200,000	4,200,000	6,100,000	-1,900,000	0.689	100.0%
2.3	Foundation RCC up to plinth	9,500,000	9,500,000	10,800,000	-1,300,000	0.880	100.0%
2.2	PCC and footings	6,800,000	6,800,000	7,400,000	-600,000	0.919	100.0%
3.1	RCC columns, ground to fourth floor	14,500,000	3,190,000	3,600,000	-410,000	0.886	22.0%
3.2	RCC beams and slabs, ground to fourth floor	21,000,000	2,940,000	3,300,000	-360,000	0.891	14.0%
2.4	Plinth beams and backfilling	3,800,000	3,800,000	4,100,000	-300,000	0.927	100.0%
1.1	Site mobilisation and enabling works	2,500,000	2,500,000	2,700,000	-200,000	0.926	100.0%
1.2	Site office, stores and labour accommodation	1,800,000	1,800,000	1,850,000	-50,000	0.973	100.0%

₹38 lakh
in three completed substructure packages — this variance is FINAL

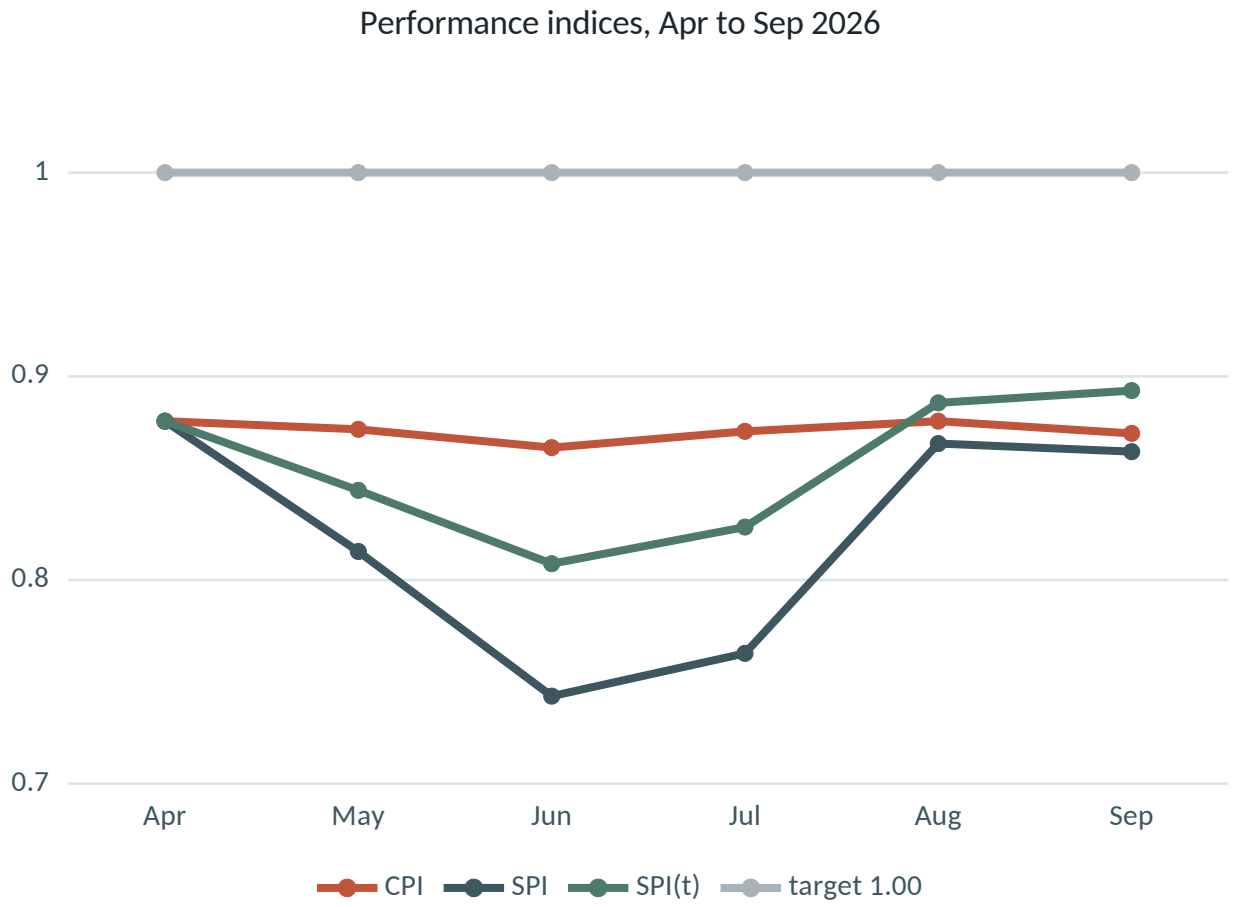
₹7.7 lakh
in the frame, 22% and 14% complete — this can still be managed

Excavation and earthwork: CPI 0.689, 100% complete.

No recovery plan touches it. Stop trying to recover the substructure, protect the frame, and tell the client ₹38 lakh is gone.

This is the slide that justifies carrying the WBS through the whole calculation instead of rolling up to one project curve.

SPI has a structural defect, and Earned Schedule fixes it



SPI = EV ÷ PV

At completion EV must equal BAC and PV must equal BAC.

So SPI returns to 1.00 on every project ever completed, however late.

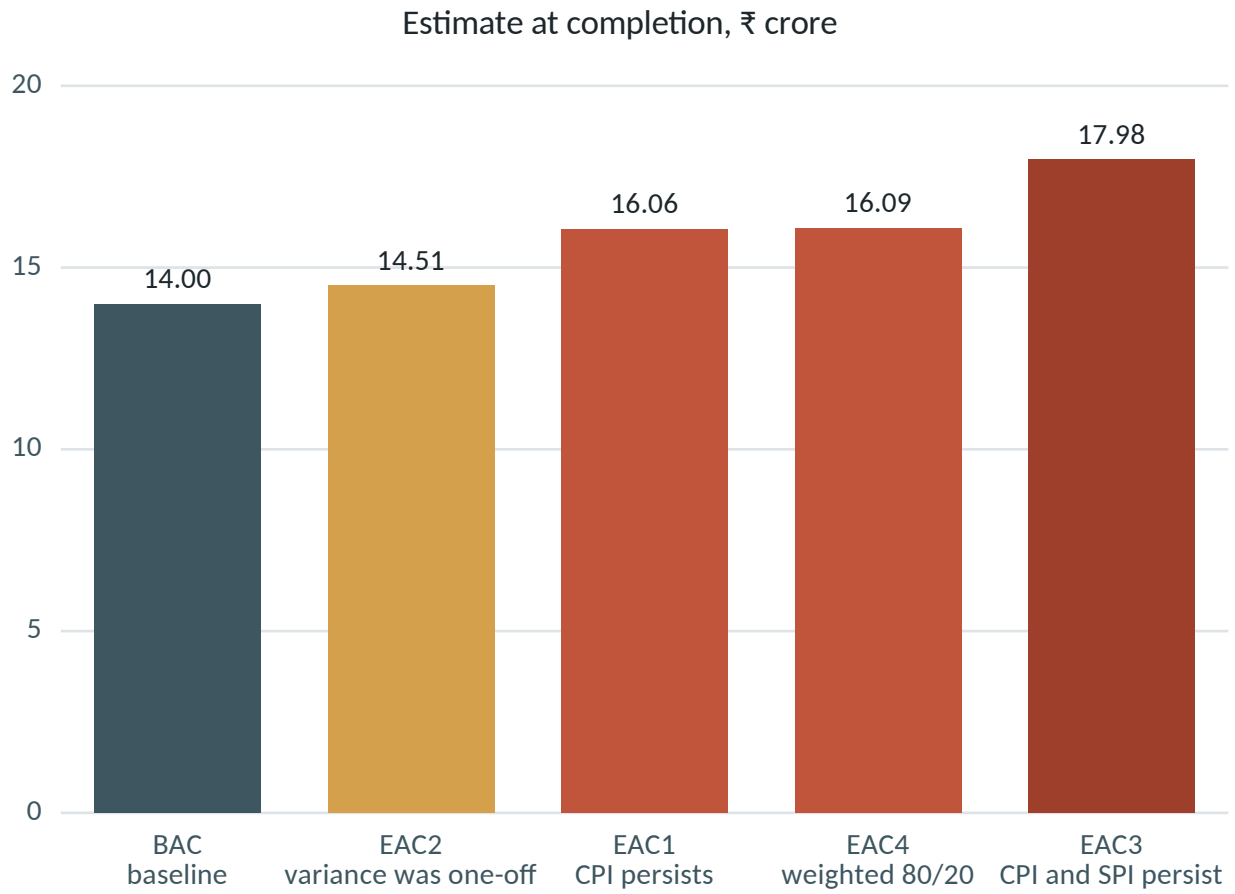
19 days behind

Earned Schedule, measured in time — SPI(t) 0.893

A site manager can act on nineteen days. Nobody can act on 0.863.

Note CPI flat at 0.86–0.88 across all six periods while SPI swings to 0.743 and back: cost performance stabilises early and rarely recovers.

The forecast is a range, and that is the honest answer



₹3.47 Cr

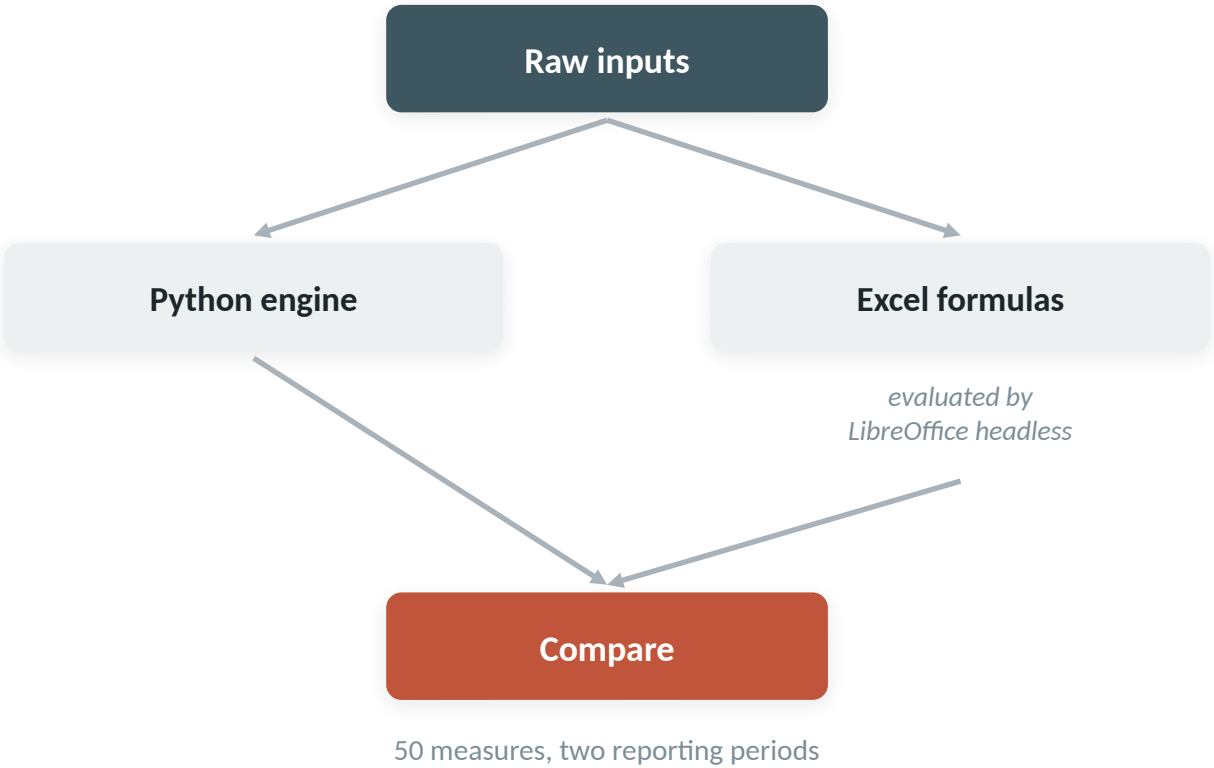
spread between the four standard methods

We deliberately do not pick one.

The four methods differ in a single assumption — what you believe about the remaining work. The spread is the method being honest, not failing.

Narrowing it needs a judgement nobody can make arithmetically: was the rock a closed event, or a symptom of a site that was never properly investigated?

How we know the numbers are right



50 measures, two reporting periods

50 / 50
comparisons agree — to ₹0.01 on currency, 1×10^{-9} on indices

150
automated tests — expected values hand-derived before the engine existed

The decision that mattered

Not to write a second Python implementation.

A Python re-implementation validates only your own understanding. Misread a formula and both sides misread it identically, the comparison passes, and both are wrong.

So the benchmark lives in a different execution environment, and it is executed — not re-implemented.

Four discrepancies, all recorded

What we found	How it resolved
EAC3 off by ₹156	**Our hand calculation was wrong, not the engine** — we rounded CPI × SPI mid-chain. Rule adopted: rounding is presentation only.
Two validation tests failing	**The tests were wrong, not the engine.** Fixing the code to make them pass would have deleted a rule that rejects earned value being double counted.
Executive summary invisible under a dark theme	**No test caught it.** Every number was correct and the tool was unusable. Found only by opening the running app.
Carry-forward routine fragile on reporting gaps	**Caught in review before it produced a wrong answer.** Clean data hid it; real data would not.

The second one matters most. The fast route to a green test suite was to change the engine until the tests passed — which would have made the project worse while making the output greener. Ask why a test fails before changing anything.

What none of this validation can do

Both implementations read the same inputs.

60% reported
40% actual

1. the engine computes a wrong EV
2. the benchmark computes the same wrong EV
3. all 50 comparisons agree
4. **every index is confidently, precisely wrong**

Earned value is percent complete × budget. Percent complete is a judgement made at site, often by someone who is paid more if it is higher. No spreadsheet and no test suite detects an error in it — only a measurement at site does.

We validated the arithmetic to nine decimal places. The arithmetic was never the risk.

And every design choice that makes this dashboard more persuasive makes the problem worse: it renders an unverifiable judgement with six-figure precision, a traffic light and a chart.

The quality of an EVM system is the quality of its percent-complete measurement.